

URUMQI, China

WMO: 514630

Lat: 43.783N

Lon: 87.617E

Elev: 3015

StdP: 13.16

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.8	-3.7	-13.1	3.0	-6.1	-8.7	3.8	-2.5	12.0	17.3	10.3	15.6	3.4	90	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.2	92.6	62.0	89.8	61.3	87.2	60.7	64.8	83.1	63.6	82.1	62.6	81.2	7.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.5	84.8	69.2	57.6	79.2	69.4	55.7	74.0	70.2	31.5	83.1	30.5	82.0	29.7	81.1	70.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.5	12.9	10.8	-11.1	98.7	4.4	3.0	-14.3	100.8	-16.8	102.6	-19.3	104.3	-22.5	106.5		
			-11.6	66.9	4.3	2.0	-14.7	68.4	-17.2	69.5	-19.6	70.7	-22.8	72.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.7	10.1	16.1	33.4	52.7	63.4	72.6	76.0	73.7	63.8	48.8	31.7	16.1
	DBStd	24.73	8.46	8.27	11.48	9.70	8.18	6.43	5.78	6.32	8.28	9.12	9.96	8.50
	HDD50	4549	1236	949	527	84	7	0	0	0	7	135	552	1052
	HDD65	7723	1701	1369	981	377	128	15	3	9	118	505	1000	1517
	CDD50	3333	0	0	11	165	422	678	805	733	420	97	2	0
	CDD65	1032	0	0	0	8	78	242	343	278	81	2	0	0
	CDH74	9327	0	0	0	76	617	2146	3318	2553	605	12	0	0
	CDH80	3337	0	0	0	12	143	735	1319	971	156	1	0	0
Wind	WSAvg	5.1	3.7	4.1	5.0	6.3	6.2	5.9	5.7	5.6	5.7	4.9	4.2	3.8
Precipitation	PrecAvg	10.5	0.4	0.4	0.7	1.3	1.4	1.3	1.1	0.9	0.9	0.9	0.8	0.6
	PrecMax	16.5	1.0	1.3	1.9	3.5	5.0	3.7	4.3	3.4	2.6	2.6	1.8	2.0
	PrecMin	5.2	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.2	0.0
	PrecStd	3.0	0.2	0.2	0.4	0.8	1.0	0.9	0.9	0.8	0.6	0.6	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.5	38.8	66.5	80.8	88.1	93.4	98.0	96.0	89.9	75.3	59.5	39.1
		MCWB	28.7	30.9	46.6	53.0	57.5	61.1	63.0	61.8	59.6	51.9	45.2	31.3
	2%	DB	28.7	33.1	60.3	75.6	83.9	90.0	93.2	92.0	84.3	69.8	52.7	33.4
		MCWB	25.2	28.8	45.2	51.3	56.3	60.4	63.0	61.6	57.6	50.3	41.5	29.1
	5%	DB	25.2	29.7	55.1	71.3	80.5	87.3	90.0	88.8	80.3	66.1	48.3	29.8
		MCWB	22.4	26.3	43.3	49.8	55.5	60.2	62.2	61.1	56.2	48.7	39.3	26.6
	10%	DB	21.3	26.9	50.2	67.5	77.2	84.4	87.1	85.6	77.1	62.3	44.4	26.9
		MCWB	19.0	24.0	41.2	48.3	54.4	59.5	61.7	60.6	55.2	47.4	37.2	24.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.5	32.4	48.7	55.5	60.0	64.7	68.0	65.7	61.6	53.8	45.7	32.6
		MCDB	32.3	36.7	62.5	74.1	80.5	82.6	87.9	81.3	82.0	69.8	57.0	36.5
	2%	WB	25.5	29.0	46.1	52.9	58.4	63.1	65.5	64.0	59.3	51.6	42.3	29.5
		MCDB	28.5	32.5	58.2	69.3	78.0	81.7	84.0	83.0	79.5	66.6	51.6	32.9
	5%	WB	22.4	26.6	43.8	51.2	57.2	62.1	64.2	62.9	57.6	49.8	39.8	26.8
		MCDB	25.1	29.8	53.8	66.9	75.8	81.0	82.7	82.1	77.3	64.2	47.4	29.5
	10%	WB	19.2	24.1	41.5	49.6	55.7	61.0	63.1	61.8	56.2	47.9	37.6	24.4
		MCDB	21.2	26.7	49.8	65.2	73.5	79.5	81.3	80.9	74.7	60.8	44.1	26.7
Mean Daily Temperature Range	5% DB	MDBR	12.0	11.7	12.7	16.6	16.7	16.2	16.2	16.9	16.9	14.8	10.7	11.3
		MCDBR	16.4	15.4	18.6	21.3	19.8	19.7	19.7	20.9	20.3	19.0	16.1	16.3
		MCWBR	13.1	11.2	9.6	8.8	7.4	6.3	6.1	6.5	7.5	8.4	9.1	12.1
	5% WB	MCDBR	16.1	14.8	17.1	18.8	17.4	16.0	16.1	16.9	18.3	17.7	15.1	14.9
		MCWBR	12.9	10.9	9.3	8.1	7.0	5.3	5.7	5.9	7.0	8.1	8.8	11.1
Clear-Sky Solar Irradiance	taub	0.333	0.386	0.444	0.498	0.488	0.498	0.479	0.460	0.435	0.416	0.371	0.333	
	taud	2.088	1.989	1.919	1.846	1.921	1.959	2.052	2.087	2.077	2.048	2.102	2.127	
	Ebn at Noon	239	245	246	244	251	249	252	252	247	231	222	226	
	Edh at Noon	33	43	53	62	59	57	52	48	45	40	32	29	
All-Sky Solar Radiation	RadAvg	636	928	1341	1746	2030	2072	2023	1860	1594	1121	696	520	
	RadStd	29	53	81	112	101	107	95	84	54	61	56	43	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.24	N/A	N/A	+236	+162	

Nomenclature: See separate page